

2(a)	<u>sum</u> / <u>total</u> momentum of bodies is constant or <u>sum</u> / <u>total</u> momentum of bodies before = <u>sum</u> / <u>total</u> momentum of bodies after	M1
	for an isolated / closed system / no (resultant) external force	A1
2(b)(i)	EPE = area under graph or $\frac{1}{2}Fx$ or $\frac{1}{2}kx^2$ and $F = kx$	C1
	energy = $\frac{1}{2} \times 12.0 \times 8.0 \times 10^{-2} = 0.48 \text{ J}$ or energy = $\frac{1}{2} \times 150 \times (8.0 \times 10^{-2})^2 = 0.48 \text{ J}$	A1
2(b)(ii)1	$4.0 v_A = 6.0 v_B$	C1
	$E_K = \frac{1}{2}mv^2$	C1
	ratio = $\frac{0.50 \times 4.0 \left(\frac{6.0}{4.0}\right)^2}{0.50 \times 6.0} = 1.5$ or ratio = $\frac{1}{1.5} \times (1.5)^2 = 1.5$	A1
2(b)(ii)2	$0.48 = E_K \text{ of A} + E_K \text{ of B}$ $= E_K \text{ of A} + (E_K \text{ of A} / 1.5) = 5/3 \times E_K \text{ of A}$	C1
	$E_K \text{ of A} = 0.29 \text{ (0.288) J}$	A1
2(b)(iii)	curve starts from origin and has decreasing gradient	M1
	final gradient of graph line is zero	A1